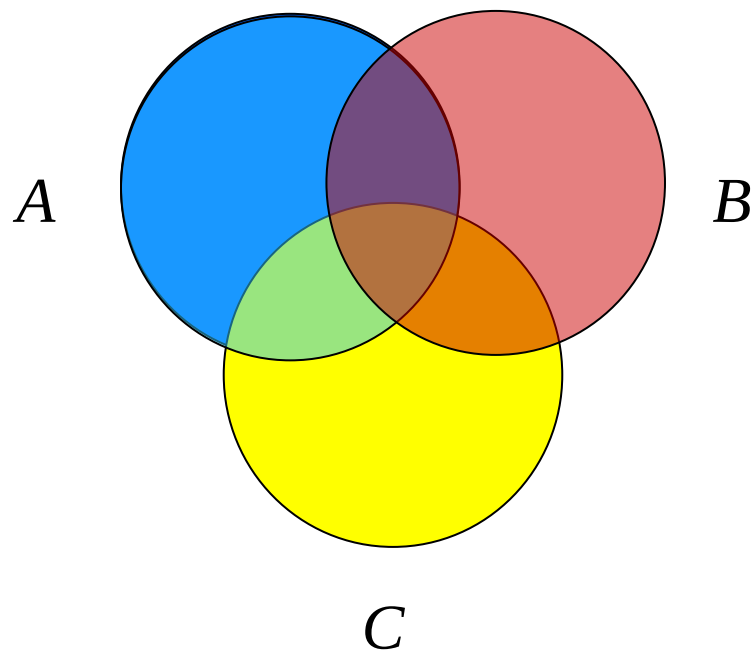


Set Theory



This Lecture

We will first introduce set theory before we do counting.

- Basic Definitions
- Operations on Sets
- Set Identities
- Russell's Paradox

Sets

A **set** is a collection of mathematical objects,
with the collection treated as a single mathematical object.

- Examples:
- real numbers, $\mathbb{R}$
 - complex numbers, $\mathbb{C}$
 - integers, $\mathbb{Z}$
 - empty set, $\emptyset$

Defining Sets

A set is just a collection of objects.

Sets can be defined directly:

e.g. $\{1, 2, 4, 8, 16, 32, \dots\}$,
 $\{\text{CSC1130}, \text{CSC2110}, \dots\}$

Order, number of occurrence are not important.

e.g. $\{A, B, C\} = \{C, B, A\} = \{A, A, B, C, B\}$

A set can be an element of another set.

$\{1, \{2\}, \{3, \{4\}\} \}$

Defining Sets by Predicates

The set of elements, x , in A such that $P(x)$ is true.

$$\{x \in A \mid P(x)\}$$

$$\{-1, 0, 1, 2, 3, 4\} = \{x \in \mathbb{Z} \mid -2 < x < 5\}$$

$$\{x \in \mathbb{R} \mid -2 < x < 5\}$$

The set of prime numbers:

$$\{p \in \mathbb{Z}^+ \mid (p > 1) \wedge (\forall a \in \mathbb{Z} \forall b \in \mathbb{Z} (a > 1 \wedge b > 1) \rightarrow a \cdot b \neq p)\}$$

Membership

The most important question in set theory is whether an element is in a set.

$$x \in A$$

x is an **element** of A

x is **in** A

Examples:

$$\pi/2 \in \{7, \text{"Albert"}, \pi/2, \mathbf{T}\}$$

$$\pi/3 \in \{7, \text{"Albert"}, \pi/2, \mathbf{T}\}$$

$$14/2 \in \{7, \text{"Albert"}, \pi/2, \mathbf{T}\}$$

$$7 \notin \{ \}$$

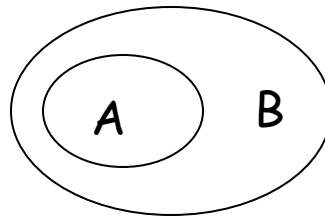
$$2/3 \notin \{ \}$$

Containment

$A \subseteq B$ A is a **subset** of B
 A is **contained in** B

Every element of A is also an element of B .

$$A \subseteq B \iff \forall x, \text{ if } x \in A \text{ then } x \in B$$



$$A \not\subseteq B \iff \exists x, x \in A \text{ and } x \notin B$$

$A \subset B$ A is a **proper subset** of B

$$A = B \iff A \subseteq B \text{ and } B \subseteq A$$

Examples

$\mathbb{R}$,

$$\{3\} \subset \{5, 7, 3\}$$

$\mathbb{P}$ $\mathbb{P}$ every set,

$$A \subseteq A$$

prime $\not\subseteq$ odd

$$\{a, \{b\}, c\} \not\subseteq \{a, b, c\}$$

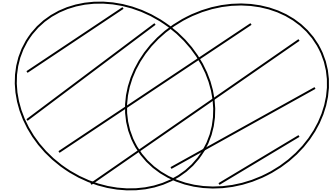
$$\{\{b\}\} \neq \{b\}$$

This Lecture

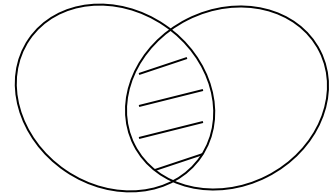
- Basic Definitions
- Operations on Sets
- Set Identities
- Russell's Paradox

Basic Operations on Sets

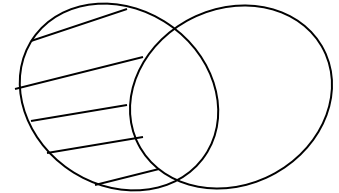
union: $A \cup B ::= \{x \mid (x \in A) \vee (x \in B)\}$



intersection: $A \cap B ::= \{x \mid x \in A \wedge x \in B\}$

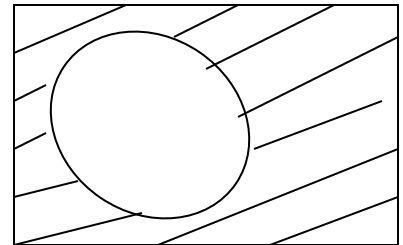


difference: $A - B ::= \{x \mid (x \in A) \wedge (x \notin B)\}$



complement: $\overline{A} ::= \{x \in D \mid x \notin A\} = D - A$

Complement can be defined only if the domain is defined



Examples

$$A = \{1, 3, 6, 8, 10\} \quad B = \{2, 4, 6, 7, 10\}$$

$$A \cap B = \{6, 10\}, \quad A \cup B = \{1, 2, 3, 4, 6, 7, 8, 10\} \quad A - B = \{1, 3, 8\}$$

$$\text{prime} \cap \text{even} = \{2\}, \quad \text{even} \cap \text{odd} = \emptyset$$

$$A = \{x \mid x = 2k \text{ for some integer } k\}, \quad B = \{x \mid x = 3k \text{ for some integer } k\}$$

$$A \cap B = \{x \mid x = 6k \text{ for some integer } k\}$$

$$A \cup B = \{x \mid x \text{ is either a multiple of 2 or a multiple of 3 (or both)}\}$$

$$A - B = \{x \mid x \text{ is even but not a multiple of 3}\}$$

Let D (i.e. the domain) be the set of integers, then

$\overline{A}$ is the set of odd numbers

Partitions of Sets

Two sets are **disjoint** if their intersection is empty.

A collection of nonempty sets $\{A_1, A_2, \dots, A_n\}$ is a **partition** of a set A if and only if

$$A = A_1 \cup A_2 \cup \dots \cup A_n$$

and $A_1, A_2, \dots, A_n$ are **mutually disjoint**.

Partitions of Sets

Two sets are **disjoint** if their intersection is empty.

A collection of nonempty sets $\{A_1, A_2, \dots, A_n\}$ is a **partition** of a set A if and only if

$$A = A_1 \cup A_2 \cup \dots \cup A_n$$

and $A_1, A_2, \dots, A_n$ are **mutually disjoint**.

$$a \equiv b \pmod{n}$$

For a positive integer n , two integers a and b are said to be **congruent modulo n** , if their difference $a - b$ is an integer multiple of n . The number n is called the **modulus** of the congruence.

$38 \equiv 14 \pmod{12}$ congruence of a and b can also be thought of as asserting that these two numbers have the same remainder after dividing by the modulus n

Partitions of Sets

Two sets are **disjoint** if their intersection is empty.

A collection of nonempty sets $\{A_1, A_2, \dots, A_n\}$ is a **partition** of a set A if and only if

$$A = A_1 \cup A_2 \cup \dots \cup A_n$$

and $A_1, A_2, \dots, A_n$ are **mutually disjoint**.

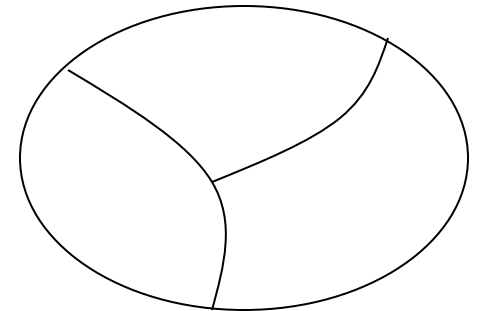
Example. Let A be the set of integers.

A_1 be the set of integers $\equiv 1 \pmod{3}$

A_2 be the set of integers $\equiv 2 \pmod{3}$

A_3 be the set of integers $\equiv 0 \pmod{3}$

$\{A_1, A_2, A_3\}$ is a partition of A



Power Sets

power set: $\text{pow}(A) ::= \{S \mid S \subseteq A\}$

$$\text{pow}(\{a,b\}) = \{\emptyset, \{a\}, \{b\}, \{a,b\}\}$$

$$\text{pow}(\{a,b,c\}) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$$

$$\begin{aligned} \text{pow}(\{a,b,c,d\}) = \{ & \emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \\ & \{a,b\}, \{a,c\}, \{b,c\}, \{a,d\}, \{b,d\}, \{c,d\}, \\ & \{a,b,c\}, \{a,b,d\}, \{a,c,d\}, \{b,c,d\}, \{a,b,c,d\} \} \end{aligned}$$

If A has n elements, then the $\text{pow}(A)$ has 2^n elements.

$P(n)$ = If A has n elements, then the $\text{pow}(A)$ has 2^n elements.

Basis: $P(0)$ is true, as the set with 0 elements has 1 subset (itself) and $2^0 = 1$.

Inductive step: Prove $P(k) \rightarrow P(k+1)$

Let T be a set such that $|T| = k+1$ and $T = \{e_1, e_2, \dots, e_k, e_{k+1}\}$.

Let $S = \{e_1, e_2, \dots, e_k\}$. Then $|S| = k$, so, S has 2^k subsets (by induction).

Note that $T = S \cup \{e_{k+1}\}$, so every subset of S is also a subset of T .

Any subset of T either contains the element e_{k+1} , or it doesn't contain e_{k+1} . If a subset of T doesn't contain e_{k+1} , then it is also a subset of S , and there are 2^k of those subsets.

On the other hand, if a subset of T does contain the element e_{k+1} , then that subset is formed by including e_{k+1} in one of the 2^k subsets of S , so T has 2^k subsets containing e_{k+1} .

We have shown that T has 2^k subsets containing e_{k+1} , and another 2^k subsets not containing e_{k+1} , so the total number of subsets of T is $2^k + 2^k =$
(2) $2^k = 2^{k+1}$

[QED]

Cartesian Products

Given two sets A and B , the Cartesian product $A \times B$ is the set of all **ordered** pairs (a,b) , where a is in A and b is in B .

Ordered pairs means the ordering is important, e.g. $(1,2) \neq (2,1)$

e.g. $A = \{x,y\}$ $B = \{1,2,3\}$

$$A \times B = \{ (x,1), (x,2), (x,3), (y,1), (y,2), (y,3) \}$$

If A has n elements and B has m elements, then $A \times B$ has mn elements.

The definition can be generalized to any number of sets.

e.g. $A = \{1,2\}$, $B = \{x,y\}$, $C = \{T,F\}$

$$A \times B \times C = \{ (1,x,T), (1,x,F), (1,y,T), (1,y,F), \\ (2,x,T), (2,x,F), (2,y,T), (2,y,F) \}$$

This Lecture

- Basic Definitions
- Operations on Sets
- Set Identities
- Russell's Paradox

Set Identities

Some basic properties of sets

$$A \cap B \subseteq A$$

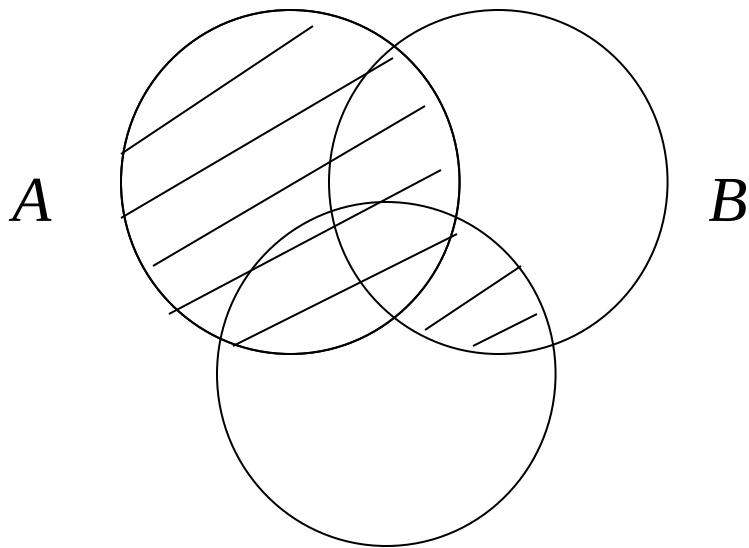
$$A \subseteq A \cup B$$

$$A \subseteq B \text{ and } B \subseteq C \rightarrow A \subseteq C$$

Set Identities

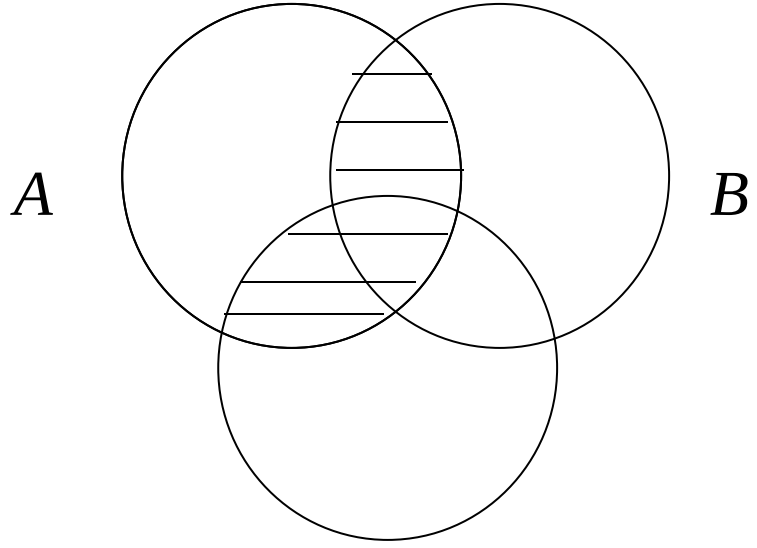
Distributive Law: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (1)

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (2)



C

(1)



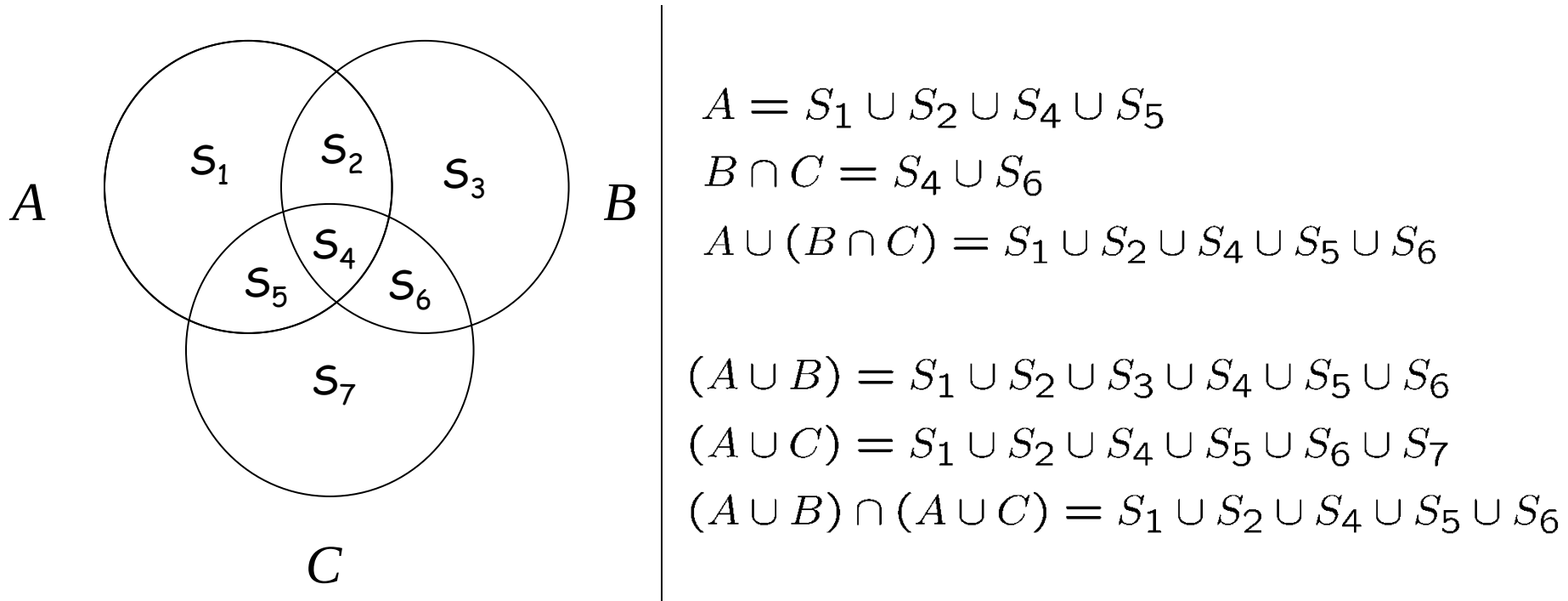
C

(2)

Set Identities

Distributive Law: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

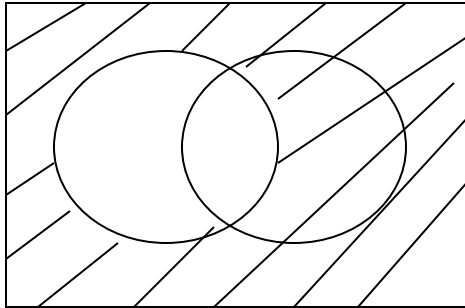
If you insist on proving this law more formally...



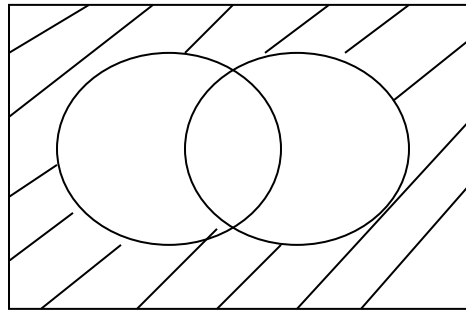
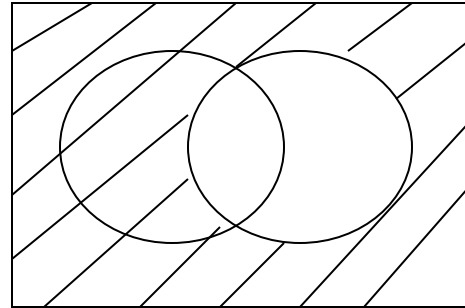
Set Identities

De Morgan's Law: $\overline{A \cup B} = \overline{A} \cap \overline{B}$

$\overline{A}$



$\overline{B}$



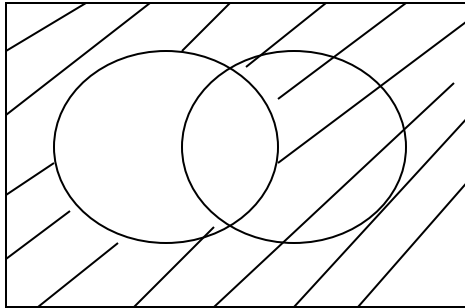
$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

Set Identities

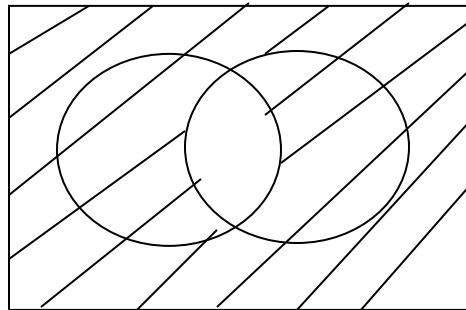
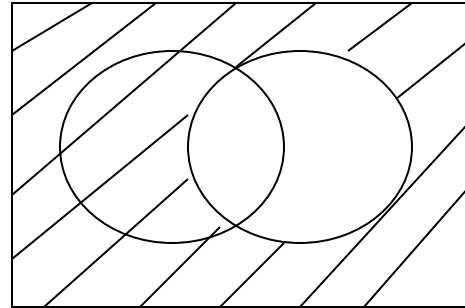
De Morgan's Law:

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

$\overline{A}$



$\overline{B}$



$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

Exercises

$$A - (A \cap B) = A - B?$$

$$(A - B) \cup (B - C) = A - C?$$

$$(A \cup B) - C = (A - C) \cup (B - C)?$$

This Lecture

- Basic Definitions
- Operations on Sets
- Set Identities
- Russell's Paradox

Russell's Paradox

Let $W ::= \{S \in \text{Sets} \mid S \notin S\}$

so $S \in W \leftrightarrow S \notin S$

Is W in W ?

Russell's Paradox

- "abnormal" set: if it is a member of itself
- "normal" set: otherwise.
- Example:
 - take the set of all squares S .
 - S is not itself a square
 - S is "normal".
 - the complementary set that contains all non-squares, S^c
 - that set is itself not a square
 - so S^c is "abnormal".

Russell's Paradox

- Now we consider the set of all normal sets, R .
- Now, try to determine whether R is normal or abnormal
- *Case 1: R is a normal set:*
 - it would be contained in the set of normal sets (itself),
 - and therefore be abnormal;
- *Case 2: R is abnormal:*
 - it would not be contained in the set of normal sets
 - So, it would not be contained in itself
 - and therefore be normal.
- This leads to the conclusion that R is neither normal nor abnormal: Russell's paradox.

Barber's Paradox

There is a male barber who shaves all those men,
and only those men,
who do not shave themselves.

Does the barber shave himself?

This Lecture

Just make sure you understand the basics of set theory.

- Basic Definitions
- Operations on Sets
- Set Identities